

BUILT — schematic Highlighter (Phase 1) + end-to-end demo suite (v0.71.0)

Click elements on vector schematics to highlight the connected net; prove the whole app hangs together with one host-side command.

☐ Highlighter — vector overlay

- `schem_overlay.py` reads the page's real geometry + text via PyMuPDF on demand (normalized 0..1, no conversion).
- `schemhl.js` renders an interactive SVG (page image + clickable paths); hover outlines, click highlights the CONNECTED net/trace (union-find on shared endpoints).
- Toggle: ☐ Highlight in the schematic viewer. Own zoom.

~45% of sheets clickable now, no conversion.

☐ Raster fallback (honest)

- Scanned sheets have no geometry — `schempaths` returns `has_vector=false`.
- The highlighter shows a clear 'this is a scan' note and the existing callout chips (NSN/part/figure -> dossier/Look-Alike) stay available.
- Flood-fill tracing for scans = Phase 2 (deferred per your choice).

No dead ends; scans use callouts.

☐ End-to-end congruence suite

- `test_routes` now starts the REAL server on the fixture and hits EVERY major route (search/collections/callouts/3D/schempaths/tags/keywords/dossier/procedure/health/...); asserts no 5xx + valid JSON.
- `RUN-ALL-TESTS.bat` = pillars + features + patterns + routes + truncation + RPS lint.
- `DEMO-SCRIPT.md`: green-gate + a 6-min end-to-end demo flow.

One button proves it works together.

HOW TO RUN THE DEMO

On Windows: (1) `RUN-ALL-TESTS.bat` -> expect ALL TESTS GREEN. (2) `run_app.bat` -> it serves + opens the page. (3) follow `docs/DEMO-SCRIPT.md`: slang search -> tag -> page/loupe/callouts -> schematic highlighter -> detailed 3D -> collections -> solve->packet -> PUB LOG dossier. Optional: `FINALIZE-OCR.bat` + `ENRICH-PUBLOG.bat` for fullest data; `run_safeguard.bat` snapshot first.

VERIFIED: `schem_overlay` on real corpus schematics (607 paths), `schemhl.js` syntax + ES5-clean, `schematics.html` toggle + both routes + expanded `test_routes` confirmed on host. CAVEAT: the sandbox mount is intermittently corrupting reads of the large `viewer_app.py`, so the server-dependent suite couldn't be RUN in-sandbox — the authoritative gate is `RUN-ALL-TESTS.bat` on Windows. Host files are intact (authoritative tools confirm).